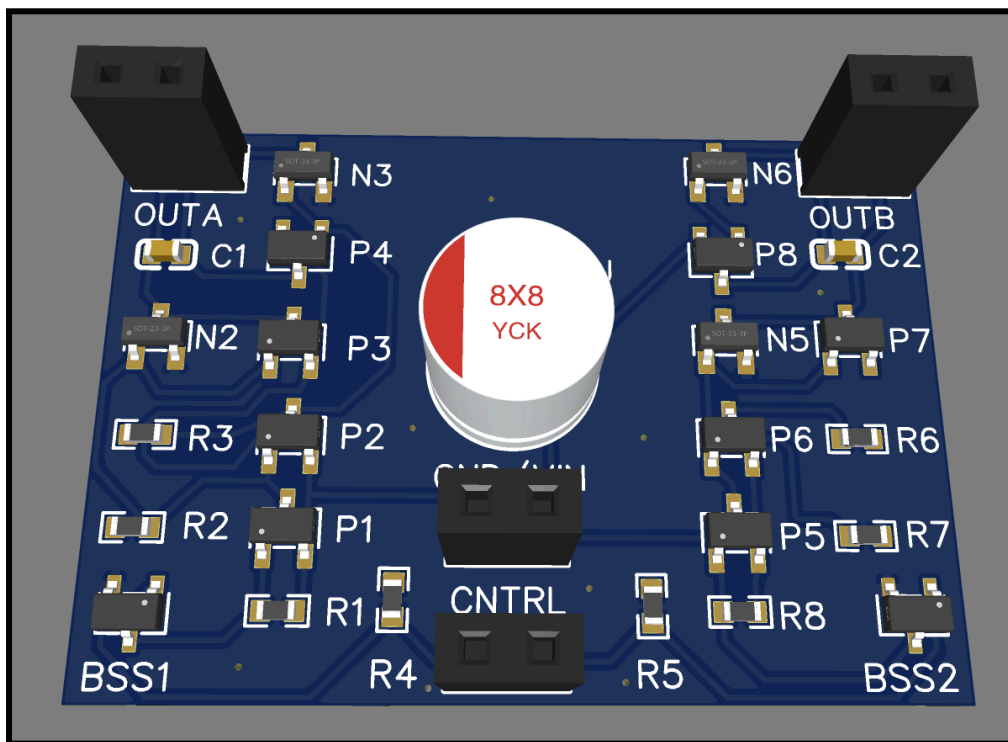




ARAM-DDHBSD_0_10-0S Arduino/Esp32 Compatible Motor Driver and Dual H-Bridge

This Dual H-Bridge is a modern MOSFET-based design from ARAM, American Robotics Assisted Manufacturing. This circuit will allow you to control the output polarity of the Dual H-Bridge using by toggling the CNTRL pin between logic Low and High. This Dual H-Bridge can support up to a 12V input and can **output up to 2.5 Amps continuously** or up to **3 Amps with the addition of a heatsink**. Amperages as high as **5 Amps surge current**. The **inbuilt logic prevents shoot-through**, it is impossible to turn both sides of the H-Bridge on simultaneously. This MOSFET-based design is **ultra efficient at 90-99%** within operating parameters. This is ideal for small microcontrollers such as **arduino or esp32**. The circuit also has a quick rise and **fall time of less than 35 nanoseconds**.





ARAM-DDHBSD_0_10-0S DATASHEET

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Standby/Quiescent (Out A & B disconnected) Current: ~ 0.008 Amps

Voltage Inputs

Voltage Input	Voltage
Max	12 Volts
Min	3 Volts* (3-5V Decreased Efficiency)

Max Switching Speed : ~ 100Khz

Logic Inputs

Logic Input	Supported
1.8V	Yes * (<i>Decreased Efficiency, Increased Rise/Fall Timing, 0-50C</i>)
3.3V	Yes
5V	Yes

Timing

Edge Timing	Time (nanoseconds)
Rise	< 35ns
Fall	< 35ns

Current Ratings (Per channel):

Current Ratings	Current (Amps)	Maximum Time (seconds)
Continuous	2.5 Amps	-
Continuous (With Heatsink)	2.5-3 Amps	-
Extended	3 Amps	5 minutes
Burst	4 Amps	60 seconds
Peak	5 Amps	30 seconds

All items tested at 18C, with no heatsink or external airflow, unless otherwise specified



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Rds(On) and Efficiency (Per Channel):

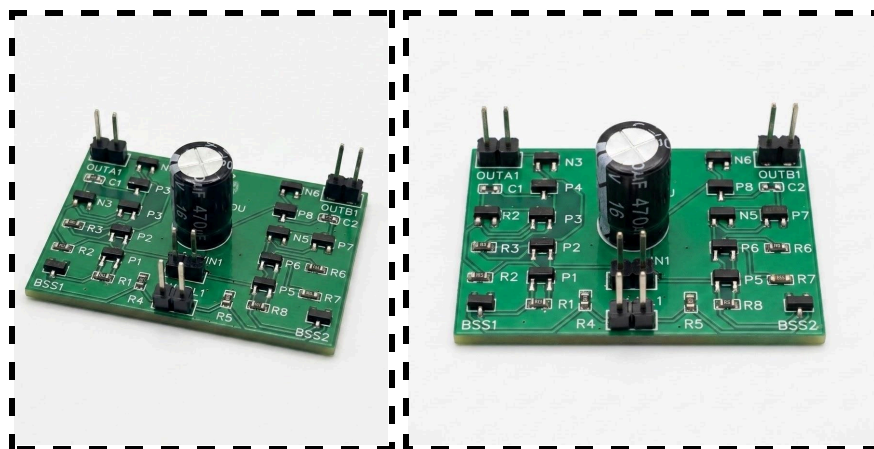
Voltage (V)	Current (Amps)	RdsOn (Ohms)	Efficiency (%)
5	1.207	0.18	95.6
5	1.623	0.18	94
5	2.274	0.18	91.4
7	0.447	0.16	99
7	1.682	0.17	95.9
7	2.253	0.18	94.2
9	0.57	0.15	99
9	2.138	0.17	95.8
9	2.831	0.18	94.2
9	3.975	0.19	91.3
11	0.689	0.16	99
11	2.565	0.18	95.8
11	3.386	0.19	94.1
11	4.491	0.21	91.2
12	0.766	0.15	99
12	2.736	0.18	95.8
12	3.574	0.19	94.1
12	4.649	0.21	91.4

All items tested at 18C, with no heatsink or external airflow, unless otherwise specified

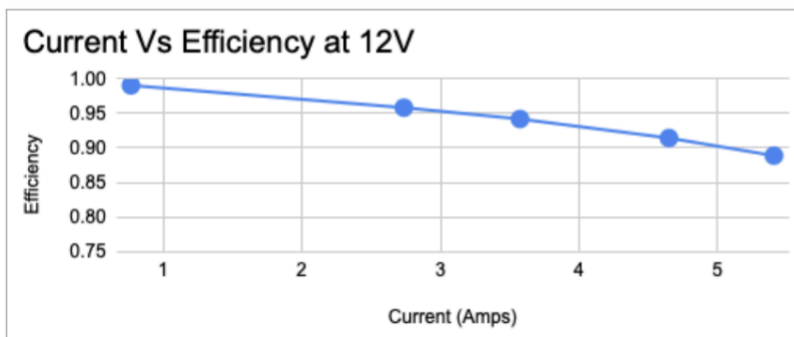
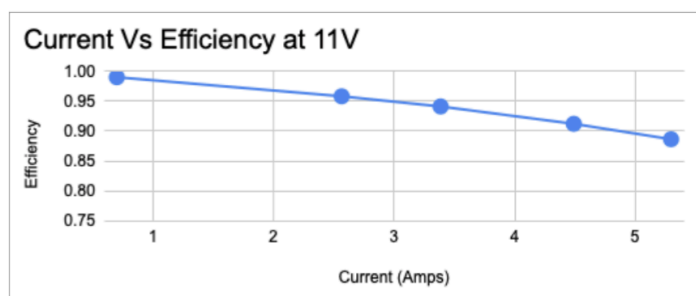
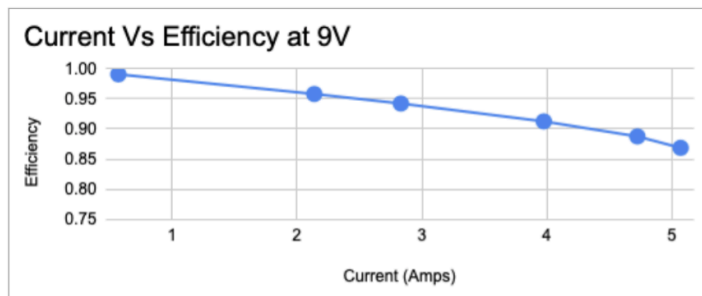
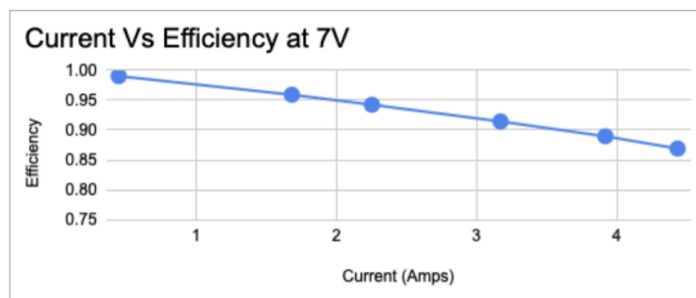
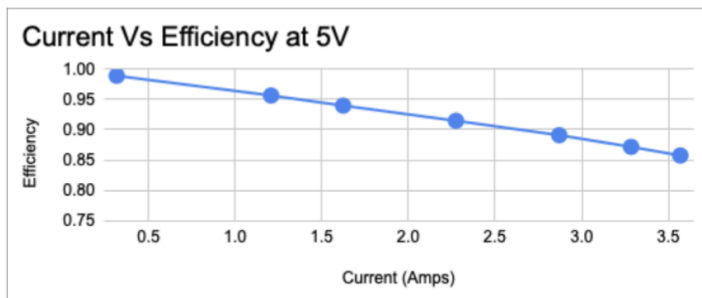


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Efficiency Vs Current (Per Channel) Graphs



(Efficiency is calculated based on $R_{ds(on)}$ characterization of the MOSFETs, verified by standard multimeter load testing.)

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Pinout Table

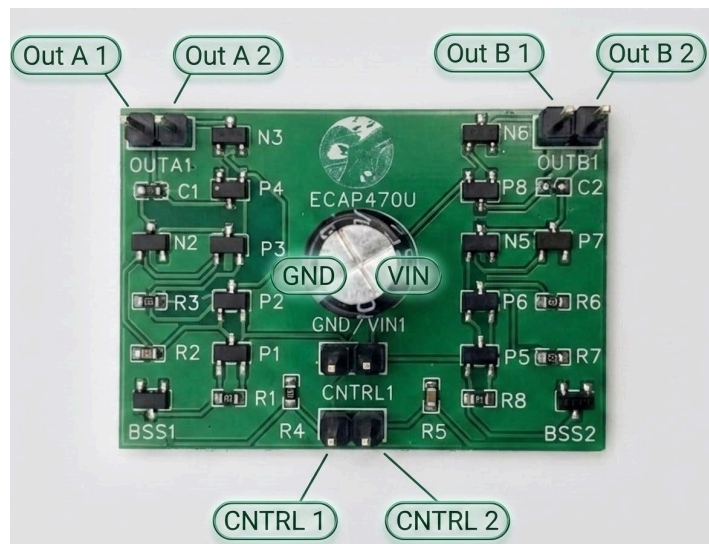
Pin Name	Use
Vin	Connect to Vin (5V-12V)
Gnd	Connect to GND (0V)
Cntrl A	Toggle between logic HIGH (1.8V*/3.3V/5V) and LOW to set the polarity of the H-Bridge. If this pin is left floating the circuit will default the logic LOW polarity.
Output A 1	Output of H-Bridge, form circuit with Output A 2
Output A 2	Output of H-Bridge, form circuit with Output A 1
Cntrl B	Toggle between logic HIGH (1.8V*/3.3V/5V) and LOW to set the polarity of the H-Bridge. If this pin is left floating the circuit will default the logic LOW polarity.
Output B 1	Output of H-Bridge, form circuit with Output B 2
Output B 2	Output of H-Bridge, form circuit with Output B 1

Physical Dimensions

28 mm x 40 mm

Weight

~8g



All items tested at 18C, with no heatsink or external airflow, unless otherwise specified



Stepper Driver Usage 1

For your convenience we provide a library written to use this motor driver. It is made for compatibility with the Arduino IDE, and usage with ESP32, Arduino, and other Arduino IDE compatible microcontrollers.

How to find the library:

1: The simplest method is to find the library by searching for the library in the Arduino IDE library manager:

In the dropdown menus select tools->Manage Libraries

In the menu that appears search "ARAM_FS_Stepper"

Select the library with the author "A.R.A.M. - American Robotics Assisted Manufacturing"

2: Alternatively you can find the library and code directly at

https://github.com/ARAM-USA-Support/ARAM_FS_Stepper

How to use the library:

Wire the board according to the pinout on the next page.

Start by declaring a stepper motor in memory:

ARAM_FS_STEPPER stepper();

Next initialize (in setup) your stepper motor:

stepper = ARAM_FS_STEPPER(CNTRL pin a, CNTRL pin b)

Where CNTRL pins a and b are the pin numbers of the microcontroller which you connect to CNTRL on the board.

Now (in loop) you're ready to use the stepper motor:

Move one step clockwise,

stepper.moveStepperCW();

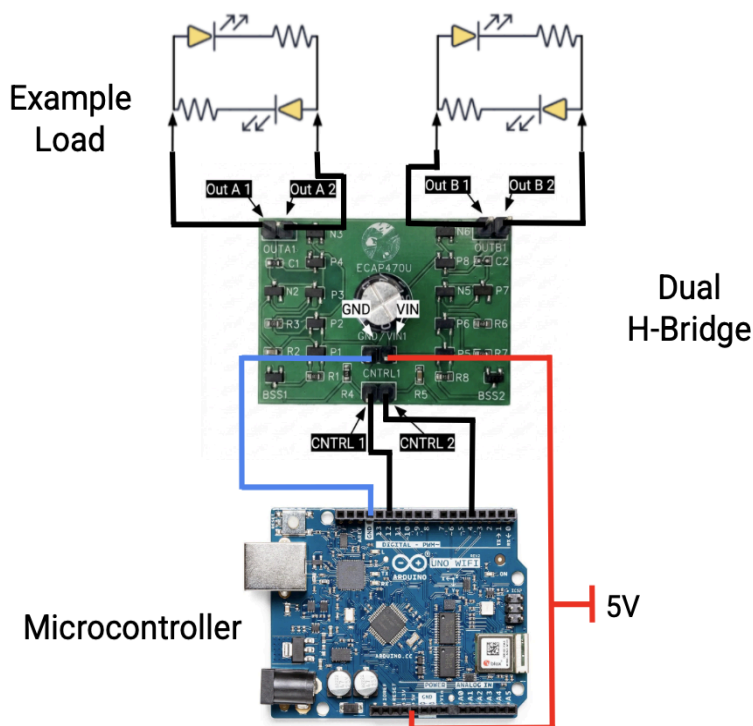
Move one step counterclockwise,

stepper.moveStepperCCW();



Stepper Driver Usage 2

Example Wiring Diagram



Example Code A:

This example shows the basic use of the ARAM_FS_STEPPER library with the motor driver. Example Code:

```
/ARAM-DDHBSD_0_13-0S      Example Code      rev: 4/29/2026
#include "ARAM_FS_Stepper.h"

#define CNTRL_PIN_A 2 // Change to your first control pin
#define CNTRL_PIN_B 1 // Change to your second control pin
#define MICROS_TO_SECONDS 1000000.0

uint32_t timeClock;
uint32_t nextSwitch;

// Set frequencies in Hz
double endFreq = 2000.0; // Target at 2kHz
double actualTimeStep = MICROS_TO_SECONDS / endFreq;

STEPPER stepper(CNTRL_PIN_A, CNTRL_PIN_B);

void setup() {
  nextSwitch = micros();
  rampEnd = startClock + (uint32_t)(RAMP_TIME * MICROS_TO_SECONDS);
}

void loop() {
  timeClock = micros();

  if(timeClock > nextSwitch){
    stepper.moveStepperCCW();
    nextSwitch = timeClock + (uint32_t)actualTimeStep;
  }
}
```



Dual H-Bridge Example Usage

Wire According The Wire Diagram Provided In The Previous Page

This example shows the basic use of a ARAM-DDHBSD_0_10-0S as a dual h-bridge.

Example Code B:

```
//ARAM-DDHBSD_0_13-0S      Example Code      rev: 4/29/2026

#define CNTRL_PIN_A 2      //change me to your desired cntrl pin A
#define CNTRL_PIN_B 1      //change me to your desired cntrl pin B
#define MICROS_TO_SECONDS (1000000)
uint32_t timeClock;
uint32_t nextSwitchA;
uint32_t nextSwitchB;
int switchAFrequency = 10; //change me to your desired switching frequency (1/seconds)
int switchBFrequency = 20; //change me to your desired switching frequency (1/seconds)
double timeStepA = MICROS_TO_SECONDS/switchAFrequency;
double timeStepB = MICROS_TO_SECONDS/switchBFrequency;
int stateA = 0;
int stateB = 0;

void setup() {
  nextSwitchB = micros();
  nextSwitchA = nextSwitchB;
  pinMode(CNTRL_PIN_A,OUTPUT);
  pinMode(CNTRL_PIN_B,OUTPUT);
}

void loop() {
  timeClock = micros();
  //every time we want to switch polarity simple shift the control pin from low to high
  //or vise versa
  //switch polarity for A
  if(timeClock > nextSwitchA){
    nextSwitchA = timeClock + timeStepA;
    if(stateA)digitalWrite(CNTRL_PIN_A,HIGH);
    else digitalWrite(CNTRL_PIN_A,LOW);

    stateA = !stateA;
  }
  //switch polarity for B
  if(timeClock > nextSwitchB){
    nextSwitchB = timeClock + timeStepB;
    if(stateB)digitalWrite(CNTRL_PIN_B,HIGH);
    else digitalWrite(CNTRL_PIN_B,LOW);

    stateB = !stateB;
  }
}
```




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Sincerely:

Jack Serlin,

Founder of American Robotics Assisted Manufacturing

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